

Kumbha Nagaswetha

Senior Data Scientist at Atlas Copco Group
M.Tech from Indian Institute of Science (IISc), Bangalore
Specialization in Deep Learning, Computer Vision

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Education

- **Indian Institute of Science, Bengaluru** 2020-2022
M.Tech in Signal Processing (Specialization in Deep Learning and Computer Vision) - CGPA : 7.0 Bengaluru, KA
- **Jawaharlal Nehru Technological University Kakinada** 2016-2020
B.Tech in Electronics and Communication Engineering - CGPA : 8.27 Kakinada, AP

Experience

- **Atlas Copco (India), Pune** 2022 - Present
Senior Data Scientist with 4 years of experience, specializing in Deep Learning, Computer Vision, and Optimization Techniques. Skilled in developing end-to-end ML pipelines for real-world applications, including machine health monitoring, survival analysis, surrogate modeling of physical processes. Proficient across deep learning architectures (CNNs, GANs, Transformers, Vision Transformers, Graph Neural Networks) and scientific machine learning methods (Neural Operators, Physics-Informed Neural Networks), with strong foundations in Signal Processing, Image Processing, Optimization, Anomaly Detection, and Domain Adaptation, with expertise in leveraging techniques like t-SNE and SHAP for model interpretability.

Projects - Deep Learning

- **Domain Adaptation in Machine Health Monitoring** Atlas Copco, June.2024–Present
 - Designed a domain adaptation approach where a model trained on source domain data was transferred and fine-tuned with minimal target domain data to detect valve damage in compressors under varying operating conditions using Vibration data.
 - Implemented Generative Adversarial methods, autoencoders, sparse autoencoders, and variational autoencoders with diverse loss functions such as mean squared error, structural similarity, cosine similarity, entropy, maximum mean discrepancy, and Mahalanobis distance.
 - Employed t-SNE and SHAP to interpret CNN-based model decisions and gain insights into learned features.
 - Achieved approximately 96% accuracy in valve damage detection with limited target training data.
- **AI based Design Optimization** Atlas Copco, November.2024–August.2025
 - Implemented the Bayesian method with multi-surrogate method for shape optimization of Jet Pump to find the best shape with least number of function evaluation and parallel computation in HPC cluster.
 - Utilized the Constant Liar approach with Kriging, RBNN, Random Forest, and SVR surrogates, achieving the best efficiency with minimal iterations. Best shape found within 30-40 iterations.
 - Built a custom library to perform Ordinary Kriging in n-dimensions, extending capabilities beyond existing implementations and enabling flexible surrogate modeling for complex design spaces.
 - Explored, compared and documented pros and cons of sampling methods (LHS, CVT, McQueen, etc.), surrogate models (Kriging, Radial Basis Functions, NNs, etc.) and optimization techniques (Bayesian, GA, PSO, BFGS) for optimization problems and used the PRESS (Predicted Residual Error Sum of Squares) method to select the best surrogate model for a given dataset.
- **Efficient Conditioning: A Theory of Pseudo-Observation Batch Bayesian Optimization** 2024–2025
 - Identified *efficient conditioning* - the ability to update surrogate predictions in closed form when data is augmented as the key structural property that makes Constant Liar (CL), Kriging Believer (KB), and fantasy models produce diverse batches under Gaussian Process surrogates while parametric surrogates (neural networks, random forests) collapse to identical batch points; formalized in Propositions 1-3 and Corollary 1 with complete proofs.
 - Derived an $\Omega(\ell)$ -scale variance suppression radius (Proposition 2) that holds for any σ -monotone acquisition function (EI, UCB, PI); unified CL, KB, and fantasy models as a single conditioning mechanism, with quantitative connections to Local Penalization (LP) and qualitative connections to Determinantal Point Processes (DPPs).
 - Introduced the Structural Diversity Diagnostic (SDD) to isolate surrogate compatibility from optimizer randomness; validated theory on Hartmann-6D, Ackley-8D, Levy-10D, and SVM hyperparameter tuning: CL/KB matches or outperforms explicit LP and is competitive with joint q-EI within noise.
 - Manuscript “Efficient Conditioning: Why Pseudo-Observation Batch Bayesian Optimization Works (When It Doesn’t)” submitted to NeurIPS 2026.
- **Speed Estimation of a Compressor from Vibration Data** Atlas Copco, Sep.2022–March.2023
 - Developed a pipeline to transform vibration signal spectra into temporal and frequency features, and designed a one-dimensional convolutional neural network integrated with LightGBM for compressor speed classification.

- Applied Principal Component Analysis (PCA), Random Forest-based and probability based feature importance to identify the most informative vibration features; also explored random forests, SVM and fully connected neural networks as baselines alongside the proposed approach.
 - Achieved 99.5% classification accuracy, with only 3 misclassified samples out of 600.
- **Time Series Forecasting and Demand Planning** Atlas Copco, August.2023–November.2023
 - Built a data-driven forecasting model to optimize sales strategy and decision-making.
 - Identified leading indicators for opportunity-to-sales conversion, achieving 87% accuracy using a Random Forest classifier.
 - Forecasted material orders using SARIMAX, ARIMA, LSTM, NN,n-hits and TCN models, achieving 90% accuracy with external indicators like inflation rate and GDP.
 - **Survival Analysis for Predictive Maintenance** Atlas Copco, April.2023– July.2023
 - Performed survival analysis to predict the life-cycle of critical components, improving maintenance planning and resource allocation.
 - Leveraged the `randomForestSRC` package in R for time-to-event predictions.
 - Developed an interactive Shiny web application for real-time data entry and visualization of survival analysis results.

Projects - Computer Vision

- **Lubricant Level Estimation Using CNN on Audio Signals** Atlas Copco, March.2023–Sep.2023
 - Led the development and implementation of the CNN architecture for Mel spectrogram analysis.
 - Explored Vision Transformers, Synthetic Minority Oversampling Technique (SMOTE), data augmentation methods such as solarization, time masking, frequency masking, Gaussian blur, and experimented with weighted convolutional networks and adaptive learning rate strategies to improve robustness.
 - Utilized Python, TensorFlow, and Keras for model training, with Librosa for audio processing and VGG16 for feature extraction; achieved an R^2 score of 0.92 on the test set and deployed a pyfunc-based pipeline with a user-friendly interface.
- **U-FNO-DA: Fourier Neural Operators for Robust Domain Adaptation** 2025–2026
 - Designed U-FNO-DA, an architecture coupling Fourier Neural Operator (FNO) blocks for spectral learning, a UNet path for multi-scale spatial context, and CORAL feature alignment for cross-condition transfer; motivated by the observation that CNN filters are implicitly calibrated to the training-time frequency resolution and harmonic amplitudes, whereas bearing fault harmonics are globally structured in frequency space.
 - Validated on the Paderborn University bearing dataset under two deployment shifts, sensor-quality (train 64 kHz, test at 32/16 kHz) and operating-condition (limited target labels, $f = 0.1\text{--}0.9$) with a calibrated mel-spectrogram protocol that preserves harmonic positions across sampling rates.
 - U-FNO-DA achieved 0 training collapses across 51 runs versus 6/51 for a controlled CNN-DA baseline under identical hyperparameters, and learned reliably with as few as 10% of target-condition labels (0.74 mean target accuracy, 3 seeds); ablation confirmed both spectral and multi-scale spatial paths are necessary.
- **HDR Image Reconstruction using Un-supervised Learning - arXiv Publication** IISc, Dec 2021 - June 2022
 - Developed an unsupervised deep learning framework for HDR reconstruction that takes multi-exposure inputs and produces HDR outputs without ground-truth supervision.
 - Explored the generation of under-exposed and over-exposed images based on brightness variations and sensor characteristics, and experimented with sparse autoencoder models for exposure fusion.
 - Introduced a novel custom loss function combining structural and perceptual quality measures, which improved visual fidelity and achieved higher MEF-SSIM scores compared to the baseline model.
- **Facial Emotion Recognition** IISc, Feb 2021 - Jun 2021
 - Implemented facial emotion recognition using JAFFE, AFED, and FER datasets to classify emotions from facial expressions.
 - Combined traditional machine learning and deep learning approaches by fine-tuning a Support Vector Machine with features extracted from a pre-trained AlexNet model.
 - Explored cross-dataset generalization and transfer learning strategies, reinforcing concepts of feature engineering and CNN-based classification.
- **A Fuzzy Based DWM filter For Impulse Noise Removal** JNTUK, Dec 2019 - Mar 2020
 - Designed a fuzzy directional weighted median filter to remove fixed-value impulse noise while preserving fine image details.

- Detected noise using fuzzy rules based on directional differences and standard deviation, and applied adaptive filtering: classical median filter in smooth regions and directional median filter in non-smooth regions aligned with the lowest variance.
- Implemented in MATLAB, achieving superior performance compared to standard median and directional weighted median filters, with significant improvements in Mean Absolute Error and Peak Signal-to-Noise Ratio, while maintaining edge structures.

- **2D Transformations for Signal Representation**

Atlas Copco, April.2023

- Converted one-dimensional vibration and audio signals into two-dimensional representations, as 2D transforms often preserve both temporal and frequency information more effectively for pattern recognition tasks.
- Explored and compared multiple transformation techniques, including spectrograms, Mel spectrograms, synchrosqueezed short-time Fourier transform (SSQ-STFT), rainbowgrams, wavelet transforms, and Hilbert–Huang transform.
- Analyzed how each representation highlighted different aspects of the data, providing insights into which methods were most suitable for downstream machine learning models such as convolutional neural networks.

- **Other Projects in IISc**

- **Scale-Invariant Feature Transform (SIFT) Analysis:** Implemented scale-space extrema detection to study keypoint robustness under rotation, scaling, blur, and noise; found partial implementation sensitive to noise and blur, highlighting the need for the full pipeline.
- **Image Classification using CNN Features:** Extracted deep features from AlexNet and ResNet for K-Nearest Neighbors classification (98.5–100% accuracy) and fine-tuned AlexNet parameters to further improve performance on custom datasets, demonstrating transfer learning effectiveness.
- **Image Segmentation using Normalized Cuts and Fully Convolutional Networks (FCN):** Compared classical graph-partitioning (N-Cut) with deep learning-based FCNs on PASCAL VOC, showing that FCNs deliver superior pixel-wise accuracy and computational efficiency.
- **Image Denoising and Sharpening:** Explored Gaussian low-pass filtering, Adaptive MMSE, SureShrink wavelet denoising, and sharpening via high-pass filtering with variable gain; adaptive methods significantly outperformed basic filtering in Mean Squared Error and perceptual quality.
- **Block Matching and 3D Filtering (BM3D):** Implemented BM3D denoising, analyzing Mean Squared Error at different stages and testing alternative filters, confirming the original Wiener-based second stage as most effective.
- **JPEG Compression Implementation:** Built the JPEG pipeline including quantization and entropy coding, evaluating trade-offs between compression efficiency and reconstructed image quality using Mean Squared Error.
- **Analysis of Perceptual Quality Metrics:** Compared MSE, SSIM, and LPIPS against human perceptual judgments, finding LPIPS more aligned with subjective quality than traditional pixel-based metrics.

Technical Skills

Technology: Machine Learning, Deep Learning, Computer Vision, Image Processing, Signal Processing, HPC, Databricks

Languages: Python, C, MATLAB, PySpark

Libraries: OpenCV, Tensorflow, Pytorch, scikit-learn, scipy, imblearn, Pyspark, Pandas, NumPy etc.

Major Courses Undertaken: Digital Image processing, Advanced Image Processing, Computer Vision, Pattern Recognition and Neural Networks.

Extra-Curricular Activities

- Participated in internship at BSNL Telecommunications in Kakinada.
- Co-ordinator of the technical event related to the department of ECE in the Annual techno-cultural fest, JNTU Kakinada, 2020.
- Attained yellow grade in Taekwondo federation of India.
- Participated in department level chess competitions.